Data Coding

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Coding data

Sometimes data needs to be coded for further use with calculations. This is particularly useful with data that deals with very small or very large numbers, or with data that needs to be classified for research purposes.

What is coding?

- Coding is a way of simplifying data to make it easier to work with
- The coding must be carried out on all values within the data set and will normally be done using a given formula
- Coding can be carried out in a number of ways:
  - Adding or subtracting a constant to each data value
  - Multiplying or dividing each data value by a constant
  - A combination of both of the above

How are statistical calculations carried out with coded data?

- If you know the **mean** or **standard deviation** of the original data it is possible to find the mean or standard deviation of the coded data and vice versa
- It is important to remember what the **mean** and **standard deviation** actually tell us about the data to understand how coding calculations work
  - The mean is a measure of **location**, changing the data set in any way will cause the mean to change in the same way

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the data set
    - Multiplying or dividing every value within the data set by a constant will change the standard deviation by the **modulus** of the constant
    - If the data were coded by multiplying or dividing by a **negative**, the standard deviation will change by the equivalent positive value
- Anytime calculations are carried out on data that has been coded,
    - The **original mean** can be found by solving the equation to reverse the coding
    - For example, if the data, x, was coded using the formula

$$y = ax + b$$

Then the mean of the coded data, $\bar{y}$ would be

$$\bar{y} = a\bar{x} + b$$

The original mean, $\bar{x}$, will be $\bar{x} = \frac{\bar{y} - b}{a}$

- The **original standard deviation**
    - Will be the same as the coded standard deviation if the data was coded by adding or subtracting a constant only
    - Can be found by reversing the coding if the data was coded by multiplying or dividing by a constant only
    - If the data was coded by a combination of both then only the multiplying or dividing will need to be reversed to find the original standard deviation
    - For example, if the data, was coded using the formula

$$y = ax + b$$

Then the standard deviation of the coded data, σ_y would be

$$\sigma_y = |a|\sigma_x$$

The original standard deviation, σ_x, will be

$$\sigma_x = \frac{\sigma_y}{|a|}$$

Worked Example

The shoulder height, h, of a group of Asian elephants living in a nature reserve are summarised in the table below.

Height, cm	Frequency, f
$200 \leq h < 220$	2
$220 \leq h < 240$	5
$240 \leq h < 260$	8

$300 \le h < 320$	2

(i) Code the data using the formula $x = \dfrac{h - 250}{20}$

(ii) Use the coded data to find an estimate for the mean and standard deviation, you may use the summary statistics $\Sigma xf = 15, \Sigma x^2 f = 59, \Sigma f = 30.$

Answer:

Find the midpoint, h, of each group:

Use the formula to code the data from the midpoints

(i)

Height, cm	Midpoint, h	$x = \dfrac{h - 250}{20}$
$200 \leqslant h < 220$	210	−2
$220 \leqslant h < 240$	230	−1
$240 \leqslant h < 260$	250	0
$260 \leqslant h < 280$	270	1
$280 \leqslant h < 300$	290	2
$300 \leqslant h < 320$	310	3

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(ii) Given the summary statistics of the coded data, x.

$\Sigma f = 30 \qquad \Sigma xf = 15 \qquad \Sigma x^2 f = 59$

Use the x values with the formula for the mean to find the coded mean, $\bar{x}$

$$\text{Coded mean} = \bar{x} = \frac{\Sigma xf}{\Sigma f} = \frac{15}{30} = 0.5$$

Reverse the coding to find the estimated mean, $\bar{h}$:

$$\bar{x} = \frac{\bar{h} - 250}{20}$$

$$20\bar{x} = \bar{h} - 250$$

$$\bar{h} = 20\bar{x} + 250$$

$$= 20\,(0.5) + 250$$

$$= 260$$

Use the x values with the formula for the standard deviation

$$\sigma x = \sqrt{\frac{\Sigma x^2 f}{\Sigma f} - \left(\frac{\Sigma xf}{\Sigma f}\right)^2} = \sqrt{\frac{59}{30} - 0.5^2} = 1.3102...$$

Only the division part needs to be reversed as the substraction will have no effect on the standard deviation.

$$\sigma x = \frac{\sigma h}{20}$$

$$\sigma h = 20\,\sigma x = 26.204$$

The mean height of the elephants, $\bar{h} = 260$ cm

The standard deviation of the elephants' height, $\sigma h = 26.2$ cm

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Examiner Tips and Tricks

- Be careful when using the formulae for the mean and standard deviation with coded summary statistics, you must make sure that you use the summary statistics consistently throughout. For example, if you use the sum of the coded data squared in the formula for the standard

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